

```
!pip install transformers torch vllm
```

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Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (4.46.3)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.5.1)
Collecting vllm
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  Downloading vllm-0.20.0-cp38-abi3-manylinux_2_35_x86_64.whl.metadata (50 kB)
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Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (3.16.1)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (0.25.1)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (24.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (2024.11.6)
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (0.20.3)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (0.12.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (0.5.1)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (4.67.1)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (4.12.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (75.8.2)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (1.13.3)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (3.4.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (3.1.4)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (2025.3.0)
Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/lib/python3.12/dist-packages (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (12.8.93)
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (12.8.90)
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (12.8.90)
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (9.10.2.21)
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /usr/local/lib/python3.12/dist-packages (12.8.4.1)
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /usr/local/lib/python3.12/dist-packages (11.3.3.83)
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /usr/local/lib/python3.12/dist-packages (10.3.9.90)
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /usr/local/lib/python3.12/dist-packages (11.7.3.90)
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /usr/local/lib/python3.12/dist-packages (12.5.8.93)
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (0.7.1)
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (3.4.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (12.8.93)
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /usr/local/lib/python3.12/dist-packages (1.13.1.3)
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (1.1)
Requirement already satisfied: cachetools in /usr/local/lib/python3.12/dist-packages (5.5.2)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (7.0.0)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.0)
Requirement already satisfied: requests>=2.26.0 in /usr/local/lib/python3.12/dist-packages (2.32.3)
Collecting blake3 (from vllm)
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  Downloading blake3-1.0.8-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (18 kB)
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Requirement already satisfied: py-cpuinfo in /usr/local/lib/python3.12/dist-packages (2.0.0)
Collecting transformers
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Downloading transformers-5.7.0-py3-none-any.whl.metadata (33 kB)
Requirement already satisfied: protobuf!=6.30.*,!=6.31.*,!=6.32.*,!=6.33.* in /usr/local/lib/python3.12/dist-packages (from transformers==5.7.0)
Requirement already satisfied: fastapi>=0.115.0 in /usr/local/lib/python3.12/dist-packages (from transformers==5.7.0)
Requirement already satisfied: aiohttp>=3.13.3 in /usr/local/lib/python3.12/dist-packages (from transformers==5.7.0)
Requirement already satisfied: openai>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from transformers==5.7.0)
Requirement already satisfied: pydantic>=2.12.0 in /usr/local/lib/python3.12/dist-packages (from transformers==5.7.0)
Requirement already satisfied: prometheus_client>=0.18.0 in /usr/local/lib/python3.12/dist-packages (from transformers==5.7.0)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (from transformers==5.7.0)
Collecting prometheus-fastapi-instrumentator>=7.0.0 (from vllm)
Downloading prometheus_fastapi_instrumentator-7.1.0-py3-none-any.whl (19 kB)
Requirement already satisfied: tiktoken>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from prometheus-fastapi-instrumentator==7.1.0)
Collecting lm-format-enforcer==0.11.3 (from vllm)
Downloading lm_format_enforcer-0.11.3-py3-none-any.whl.metadata (17 kB)
Collecting llguidance<1.4.0,>=1.3.0 (from vllm)
Downloading llguidance-1.3.0-cp39-abi3-manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl (1.1 MB)
Collecting outlines_core==0.2.14 (from vllm)
Downloading outlines_core-0.2.14-cp312-cp312-manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl (1.1 MB)
Collecting diskcache==5.6.3 (from vllm)
Downloading diskcache-5.6.3-py3-none-any.whl.metadata (20 kB)
Collecting lark==1.2.2 (from vllm)
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Collecting xgrammar<1.0.0,>=0.1.32 (from vllm)
Downloading xgrammar-0.1.34-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_5_x86_64.whl (1.1 MB)
Collecting partial-json-parser (from vllm)
Downloading partial_json_parser-0.2.1.1.post7-py3-none-any.whl.metadata (5.4 kB)
Requirement already satisfied: pyzmq>=25.0.0 in /usr/local/lib/python3.12/dist-packages (from partial-json-parser)
Collecting msgspec (from vllm)
Downloading msgspec-0.21.1-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl (1.1 MB)
Collecting gguf>=0.17.0 (from vllm)
Downloading gguf-0.18.0-py3-none-any.whl.metadata (4.5 kB)
Collecting mistral_common>=1.11.0 (from mistral_common[image]>=1.11.0->mistral_common)
Downloading mistral_common-1.11.1-py3-none-any.whl.metadata (5.9 kB)
Requirement already satisfied: opencv-python-headless>=4.13.0 in /usr/local/lib/python3.12/dist-packages (from mistral_common[image]>=1.11.0->mistral_common)
Requirement already satisfied: six>=1.16.0 in /usr/local/lib/python3.12/dist-packages (from mistral_common[image]>=1.11.0->mistral_common)
Collecting setuptools (from torch)
Downloading setuptools-80.10.2-py3-none-any.whl.metadata (6.6 kB)
Requirement already satisfied: einops in /usr/local/lib/python3.12/dist-packages (from torch)
Collecting compressed-tensors==0.15.0.1 (from vllm)
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Collecting depyf==0.20.0 (from vllm)
Downloading depyf-0.20.0-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: cloudpickle in /usr/local/lib/python3.12/dist-packages (from depyf==0.20.0)
Requirement already satisfied: watchfiles in /usr/local/lib/python3.12/dist-packages (from depyf==0.20.0)
Requirement already satisfied: python-json-logger in /usr/local/lib/python3.12/dist-packages (from depyf==0.20.0)
Collecting ninja (from vllm)
Downloading ninja-1.13.0-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl (1.1 MB)
Collecting pybase64 (from vllm)
Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl (1.1 MB)
Collecting cbor2 (from vllm)
Downloading cbor2-6.0.1-cp312-cp312-manylinux_2_28_x86_64.whl.metadata (3.1 kB)

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Collecting ijson (from vllm)
  Downloading ijson-3.5.0-cp312-cp312-manylinux2014_x86_64.manylinux_2_
Collecting setproctitle (from vllm)
  Downloading setproctitle-1.3.7-cp312-cp312-manylinux1_x86_64.manylinu
Collecting openai-harmony>=0.0.3 (from vllm)
  Downloading openai_harmony-0.0.8-cp38-abi3-manylinux_2_17_x86_64.many
Collecting anthropic>=0.71.0 (from vllm)
  Downloading anthropic-0.97.0-py3-none-any.whl.metadata (3.1 kB)
Collecting model-hosting-container-standards<1.0.0,>=0.1.13 (from vllm)
  Downloading model_hosting_container_standards-0.1.14-py3-none-any.whl
Requirement already satisfied: mcp in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: opentelemetry-sdk>=1.27.0 in /usr/local/
Requirement already satisfied: opentelemetry-api>=1.27.0 in /usr/local/
Collecting opentelemetry-exporter-otlp>=1.27.0 (from vllm)
  Downloading opentelemetry_exporter_otlp-1.41.1-py3-none-any.whl.metac
Collecting opentelemetry-semantic-conventions-ai>=0.4.1 (from vllm)
  Downloading opentelemetry_semantic_conventions_ai-0.5.1-py3-none-any.
Collecting numba==0.65.0 (from vllm)
  Downloading numba-0.65.0-cp312-cp312-manylinux2014_x86_64.manylinux_2
Collecting torch
  Downloading torch-2.11.0-cp312-cp312-manylinux_2_28_x86_64.whl.metada
Collecting torchaudio==2.11.0 (from vllm)
  Downloading torchaudio-2.11.0-cp312-cp312-manylinux_2_28_x86_64.whl.m
Collecting torchvision==0.26.0 (from vllm)
  Downloading torchvision-0.26.0-cp312-cp312-manylinux_2_28_x86_64.whl.
Collecting flashinfer-python==0.6.8.post1 (from vllm)
  Downloading flashinfer_python-0.6.8.post1-py3-none-any.whl.metadata (
Collecting flashinfer-cubin==0.6.8.post1 (from vllm)
  Downloading flashinfer_cubin-0.6.8.post1-py3-none-any.whl.metadata (1
Collecting apache-tvm-ffi==0.1.9 (from vllm)
  Downloading apache_tvm_ffi-0.1.9-cp312-abi3-manylinux_2_24_x86_64.mar
Collecting tilelang==0.1.9 (from vllm)
  Downloading tilelang-0.1.9-cp38-abi3-manylinux_2_27_x86_64.manylinux_
Collecting nvidia-cudnn-frontend<1.19.0,>=1.13.0 (from vllm)
  Downloading nvidia_cudnn_frontend-1.18.0-cp312-cp312-manylinux_2_27_x
Collecting fastsafetensors>=0.2.2 (from vllm)
  Downloading fastsafetensors-0.3-cp312-cp312-manylinux_2_27_x86_64.mar
Collecting nvidia-cutlass-dsl>=4.4.2 (from vllm)
  Downloading nvidia_cutlass_dsl-4.4.2-py3-none-any.whl.metadata (2.7 k
Collecting quack-kernels>=0.3.3 (from vllm)
  Downloading quack_kernels-0.4.0-py3-none-any.whl.metadata (640 bytes)
Collecting cuda-toolkit==13.0.2 (from cuda-toolkit[cublas,cudart,cufft,
  Downloading cuda_toolkit-13.0.2-py2.py3-none-any.whl.metadata (9.4 kF
Collecting cuda-bindings<14,>=13.0.3 (from torch)
  Downloading cuda_bindings-13.2.0-cp312-cp312-manylinux_2_24_x86_64.ma
Collecting nvidia-cudnn-cu13==9.19.0.56 (from torch)
  Downloading nvidia_cudnn_cu13-9.19.0.56-py3-none-manylinux_2_27_x86_6
Collecting nvidia-cusparselt-cu13==0.8.0 (from torch)
  Downloading nvidia_cusparselt_cu13-0.8.0-py3-none-manylinux2014_x86_6
Collecting nvidia-nccl-cu13==2.28.9 (from torch)
  Downloading nvidia_nccl_cu13-2.28.9-nv3-none-manylinux_2_18_x86_64.wh

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Collecting nvidia-nvshmem-cu13==3.4.5 (from torch)
  Downloading nvidia_nvshmem_cu13-3.4.5-py3-none-manylinux2014_x86_64.whl (1.1 MB)
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Collecting nvidia-cublas==13.1.0.3.* (from cuda-toolkit[cublas,cudart,cuda-rt,cuda-ml-c]
  Downloading nvidia_cublas-13.1.0.3-py3-none-manylinux_2_27_x86_64.whl (1.1 MB)
Collecting nvidia-cuda-runtime==13.0.96.* (from cuda-toolkit[cublas,cudart,cuda-rt,cuda-ml-c]
  Downloading nvidia_cuda_runtime-13.0.96-py3-none-manylinux2014_x86_64.whl (1.1 MB)
Collecting nvidia-cufft==12.0.0.61.* (from cuda-toolkit[cublas,cudart,cuda-rt,cuda-ml-c]
  Downloading nvidia_cufft-12.0.0.61-py3-none-manylinux2014_x86_64.whl (1.1 MB)
Collecting nvidia-cufile==1.15.1.6.* (from cuda-toolkit[cublas,cudart,cuda-rt,cuda-ml-c]
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Collecting nvidia-cuda-cupti==13.0.85.* (from cuda-toolkit[cublas,cudart,cuda-rt,cuda-ml-c]
  Downloading nvidia_cuda_cupti-13.0.85-py3-none-manylinux_2_25_x86_64.whl (1.1 MB)
Collecting nvidia-curand==10.4.0.35.* (from cuda-toolkit[cublas,cudart,cuda-rt,cuda-ml-c]
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Collecting nvidia-cusolver==12.0.4.66.* (from cuda-toolkit[cublas,cudart,cuda-rt,cuda-ml-c]
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  Downloading nvidia_nvjitlink-13.0.88-py3-none-manylinux2010_x86_64.whl (1.1 MB)
Collecting nvidia-cuda-nvrtc==13.0.88.* (from cuda-toolkit[cublas,cudart,cuda-rt,cuda-ml-c]
  Downloading nvidia_cuda_nvrtc-13.0.88-py3-none-manylinux2010_x86_64.whl (1.1 MB)
Collecting nvidia-nvtx==13.0.85.* (from cuda-toolkit[cublas,cudart,cufft,cuda-rt,cuda-ml-c]
  Downloading nvidia_nvtx-13.0.85-py3-none-manylinux1_x86_64.whl (1.1 MB)
Collecting astor (from depyf==0.20.0->vllm)
  Downloading astor-0.8.1-py2.py3-none-any.whl.metadata (4.2 kB)
Requirement already satisfied: dill in /usr/local/lib/python3.12/dist-packages (from astor)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from astor)
Collecting cuda-tile (from flashinfer-python==0.6.8.post1->vllm)
  Downloading cuda_tile-1.3.0-cp312-cp312-manylinux2014_x86_64.whl.metadata (1.1 kB)
Requirement already satisfied: nvidia-ml-py in /usr/local/lib/python3.12/dist-packages (from cuda-tile)
Requirement already satisfied: tabulate in /usr/local/lib/python3.12/dist-packages (from cuda-tile)
Collecting interegular>=0.3.2 (from lm-format-enforcer==0.11.3->vllm)
  Downloading interegular-0.3.3-py37-none-any.whl.metadata (3.0 kB)
Collecting llvmlite<0.48,>=0.47.0dev0 (from numba==0.65.0->vllm)
  Downloading llvmlite-0.47.0-cp312-cp312-manylinux2014_x86_64.whl (1.1 MB)
Collecting torch-c-dlpack-ext (from tilelang==0.1.9->vllm)
  Downloading torch_c_dlpack_ext-0.1.5-cp312-cp312-manylinux_2_24_x86_64.whl (1.1 MB)
Requirement already satisfied: ml-dtypes in /usr/local/lib/python3.12/dist-packages (from torch-c-dlpack-ext)
Collecting z3-solver<4.15.5,>=4.13.0 (from tilelang==0.1.9->vllm)
  Downloading z3_solver-4.15.4.0-py3-none-manylinux_2_17_x86_64.whl (1.1 MB)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from z3-solver)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from z3-solver)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from z3-solver)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from z3-solver)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from z3-solver)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from z3-solver)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from z3-solver)

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Requirement already satisfied: yarll<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (1.17.0)
Requirement already satisfied: anyio<5,>=3.5.0 in /usr/local/lib/python3.12/dist-packages (3.5.0)
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (1.7.0)
Requirement already satisfied: docstring-parser<1,>=0.15 in /usr/local/lib/python3.12/dist-packages (0.15)
Requirement already satisfied: httpx<1,>=0.25.0 in /usr/local/lib/python3.12/dist-packages (0.25.0)
Requirement already satisfied: jiter<1,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (0.4.0)
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (1.3.1)
Requirement already satisfied: starlette<0.47.0,>=0.46.0 in /usr/local/lib/python3.12/dist-packages (0.46.0)
Requirement already satisfied: typing-inspection<0.10.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (0.4.2)
Requirement already satisfied: annotated-doc<0.2.0,>=0.0.2 in /usr/local/lib/python3.12/dist-packages (0.0.2)
Collecting fastapi-cli<0.0.10,>=0.0.8 (from fastapi-cli[standard]>=0.0.8; extra==standard)
  Downloading fastapi_cli-0.0.24-py3-none-any.whl.metadata (6.4 kB)
Collecting fastar<0.12.0,>=0.9.0 (from fastapi[standard]>=0.115.0->vllm)
  Downloading fastar-0.11.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (1.5 kB)
Requirement already satisfied: python-multipart<0.0.19,>=0.0.18 in /usr/local/lib/python3.12/dist-packages (0.0.18)
Collecting email-validator<3.0.0,>=2.0.0 (from fastapi[standard]>=0.115.0->vllm)
  Downloading email_validator-2.3.0-py3-none-any.whl.metadata (26 kB)
Requirement already satisfied: uvicorn<0.30.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (0.29.0)
Requirement already satisfied: pydantic-settings<2.5.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (2.0.0)
Collecting pydantic-extra-types<2.10.0,>=2.0.0 (from fastapi[standard]>=0.115.0->vllm)
  Downloading pydantic_extra_types-2.11.1-py3-none-any.whl.metadata (4.1 kB)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (1.4.3)
Requirement already satisfied: MarkupSafe<3.0,>=2.0 in /usr/local/lib/python3.12/dist-packages (2.0.1)
Requirement already satisfied: jsonschema<4.22.0,>=4.21.1 in /usr/local/lib/python3.12/dist-packages (4.21.1)
Collecting jmespath<1.0.0,>=0.10.0 (from model-hosting-container-standards<1.0.0,>=0.10.0; extra==standard)
  Downloading jmespath-1.1.0-py3-none-any.whl.metadata (7.6 kB)
Collecting supervisor<4.3.0,>=4.2.0 (from model-hosting-container-standards<1.0.0,>=0.10.0; extra==standard)
  Downloading supervisor-4.3.0-py2.py3-none-any.whl.metadata (87 kB)
 87.9/87.9 kB 6.0 MB/s eta 0:00
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Collecting opentelemetry-exporter-otlp-proto-grpc==1.41.1 (from opentelemetry-exporter-otlp-proto-grpc[standard]==1.41.1; extra==standard)
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Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/
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_____ 90.2/90.2 MB 9.3 MB/s eta 0
Downloading nvidia_cuda_runtime-13.0.96-py3-none-manylinux2014_x86_64.n
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[illegible]

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Attempting uninstall: llvmlite
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Attempting uninstall: cuda-python
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Attempting uninstall: torch
Found existing installation: torch 2.10.0+cu128
Uninstalling torch-2.10.0+cu128:
  Successfully uninstalled torch-2.10.0+cu128
Attempting uninstall: opentelemetry-exporter-otlp-proto-http
Found existing installation: opentelemetry-exporter-otlp-proto-http 1.38.0
Uninstalling opentelemetry-exporter-otlp-proto-http-1.38.0:
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Attempting uninstall: transformers
Found existing installation: transformers 5.0.0
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  Successfully uninstalled transformers-5.0.0

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Successfully uninstalled transformers-3.0.0
Attempting uninstall: torchvision
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ipython 7.34.0 requires jedi>=0.16, which is not installed.
google-adk 1.29.0 requires opentelemetry-api<1.39.0,>=1.36.0, but you h
google-adk 1.29.0 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you h
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libraft-cu12 26.2.0 requires cuda-toolkit[cublas,curand,cusolver,cuspar
Successfully installed anthropic-0.97.0 apache-tvm-ffi-0.1.9 astor-0.8.
WARNING: The following packages were previously imported in this
runtime:
  [_distutils_hack,cuda]
You must restart the runtime in order to use newly installed versions.

```

RESTART SESSION

```

import torch
import vllm
print("✅ vLLM is loaded and ready!")
print(f"✅ GPU being used: {torch.cuda.get_device_name(0)}")

```

✅ vLLM is loaded and ready!
 ✅ GPU being used: Tesla T4

```

import time
import torch
import gc
from transformers import AutoModelForCausalLM, AutoTokenizer
from vllm import LLM, SamplingParams

# Small model for safe testing on Colab T4 or smaller GPUs
MODEL_ID = "TinyLlama/TinyLlama-1.1B-Chat-v1.0"

```

```

# Simulating multiple users sending requests at once
prompts = [
    "The future of artificial intelligence is",
    "To bake a chocolate cake, you need",
    "Once upon a time in a distant galaxy",
    "The most important invention of the 20th century was",
    "In order to learn Python effectively, one should",
    "The quick brown fox jumps over the",
    "Quantum computing will change the world because",
    "The best way to manage memory in C is"
]

MAX_TOKENS = 100
BATCH_SIZE = len(prompts)

print("=" * 50)
print(" EXPERIMENT 1: HUGGINGFACE (STANDARD INFERENCE)")
print("=" * 50)

# -----
# HUGGINGFACE BASELINE TEST
# -----

# Load tokenizer
tokenizer = AutoTokenizer.from_pretrained(MODEL_ID)

# Add padding token if missing (needed for batch processing)
if tokenizer.pad_token is None:
    tokenizer.pad_token = tokenizer.eos_token

# Load model using safer auto device placement
hf_model = AutoModelForCausalLM.from_pretrained(
    MODEL_ID,
    torch_dtype=torch.float16,
    device_map="auto"
)

# Convert prompts into model input tensors
inputs = tokenizer(
    prompts,
    return_tensors="pt",
    padding=True
).to("cuda")

```

```

# Count original input tokens
input_token_count = inputs["input_ids"].numel()

# Reset GPU memory tracker
torch.cuda.reset_peak_memory_stats()

# Start timer
start_time = time.time()

# Run generation using standard HuggingFace inference
with torch.no_grad():
    hf_outputs = hf_model.generate(
        **inputs,
        max_new_tokens=MAX_TOKENS,
        use_cache=True, # Standard contiguous KV-cache
        pad_token_id=tokenizer.pad_token_id
    )

# IMPORTANT: wait for GPU to finish before stopping timer
torch.cuda.synchronize()
hf_time = time.time() - start_time

# Track approximate peak GPU memory usage
hf_memory = torch.cuda.max_memory_allocated() / (1024 ** 3)

# Count generated tokens only
hf_generated_tokens = hf_outputs.numel() - input_token_count

# Throughput = tokens generated per second
hf_throughput = hf_generated_tokens / hf_time

print(f"HuggingFace Latency: {hf_time:.2f} seconds")
print(f"HuggingFace Throughput: {hf_throughput:.2f} tokens/sec")
print(f"HuggingFace Peak VRAM: {hf_memory:.2f} GB\n")

# -----
# CLEAN GPU MEMORY BEFORE vLLM
# -----

print("Cleaning up VRAM...")

del hf_model
del inputs
del hf_outputs

```

```

gc.collect()
torch.cuda.empty_cache()

# Give CUDA a moment to fully release memory
time.sleep(2)

print("=" * 50)
print(" EXPERIMENT 2: vLLM (PAGEDATTENTION)")
print("=" * 50)

# -----
# vLLM PAGEDATTENTION TEST
# -----

# Restrict memory usage so Colab T4 doesn't crash
vllm_model = LLM(
    model=MODEL_ID,
    gpu_memory_utilization=0.4
)

sampling_params = SamplingParams(
    max_tokens=MAX_TOKENS
)

# Reset memory tracker again
torch.cuda.reset_peak_memory_stats()

# Start timer
start_time = time.time()

# Run vLLM generation
vllm_outputs = vllm_model.generate(
    prompts,
    sampling_params
)

# IMPORTANT: wait for GPU to finish
torch.cuda.synchronize()
vllm_time = time.time() - start_time

# Approximate execution memory footprint
vllm_memory = torch.cuda.max_memory_allocated() / (1024 ** 3)

# Count total generated tokens
vllm_generated_tokens = sum(
    len(output.outputs[0].tokens)

```

```

        len(output.outputs[0].token_ids)
        for output in vllm_outputs
    )

# Throughput calculation
vllm_throughput = vllm_generated_tokens / vllm_time

print(f"vLLM Latency: {vllm_time:.2f} seconds")
print(f"vLLM Throughput: {vllm_throughput:.2f} tokens/sec")
print(f"vLLM Approx. Peak VRAM: {vllm_memory:.2f} GB\n")

print("=" * 50)
print(" RESULTS SUMMARY")
print("=" * 50)

print(f"Throughput Increase: {vllm_throughput / hf_throughput:.2f}x faster")
print(f"Latency Improvement: {hf_time / vllm_time:.2f}x faster response")

```

```

=====
EXPERIMENT 1: HUGGINGFACE (STANDARD INFERENCE)
=====
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your setti
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to ac
warnings.warn(
config.json: 100% 608/608 [00:00<00:00, 13.1kB/s]

tokenizer_config.json: 1.29k/? [00:00<00:00, 32.4kB/s]

tokenizer.model: 100% 500k/500k [00:00<00:00, 2.50MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenti
tokenizer.json: 1.84M/? [00:00<00:00, 11.0MB/s]

special_tokens_map.json: 100% 551/551 [00:00<00:00, 8.91kB/s]
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!
model.safetensors: 100% 2.20G/2.20G [00:23<00:00, 170MB/s]

Loading weights: 100% 201/201 [00:03<00:00, 58.98it/s]

generation_config.json: 100% 124/124 [00:00<00:00, 2.56kB/s]
[transformers] A decoder-only architecture is being used, but right-pac
[transformers] Both `max_new_tokens` (=100) and `max_length` (=2048) see
HuggingFace Latency: 4.91 seconds
HuggingFace Throughput: 162.86 tokens/sec

```

```
-----
HuggingFace Peak VRAM: 2.08 GB
```

```
Cleaning up VRAM...
```

```
=====
EXPERIMENT 2: vLLM (PAGEDATTENTION)
=====
INFO 04-29 19:11:08 [utils.py:233] non-default args: {'gpu_memory_utili
INFO 04-29 19:11:35 [nixl_utils.py:20] Setting UCX_RCACHE_MAX_UNRELEASED
WARNING 04-29 19:11:35 [nixl_utils.py:34] NIXL is not available
WARNING 04-29 19:11:35 [nixl_utils.py:44] NIXL agent config is not avail
INFO 04-29 19:11:35 [model.py:555] Resolved architecture: LlamaForCausalLM
WARNING 04-29 19:11:35 [model.py:1965] Your device 'Tesla T4' (with compute
WARNING 04-29 19:11:35 [model.py:2018] Casting torch.bfloat16 to torch.float16
INFO 04-29 19:11:35 [model.py:1680] Using max model len 2048
INFO 04-29 19:11:35 [scheduler.py:239] Chunked prefill is enabled with
INFO 04-29 19:11:35 [vllm.py:840] Asynchronous scheduling is enabled.
INFO 04-29 19:11:35 [kernel.py:205] Final IR op priority after setting
WARNING 04-29 19:11:35 [system_utils.py:157] We must use the `spawn` method
```

```
Rendering prompts: 100% 8/8 [00:00<00:00, 6.88it/s]
```

```
Processed prompts: 100%
```

```
8/8 [00:01<00:00, 1.71s/it, est. speed input: 48.33 toks/s, output: 465.79 toks/s]
```

```
vLLM Latency: 1.88 seconds
vLLM Throughput: 425.28 tokens/sec
vLLM Approx. Peak VRAM: 0.01 GB
```

```
=====
RESULTS SUMMARY
=====
Throughput Increase: 2.61x faster with vLLM
```

```
results_text = f"""
EXPERIMENTAL RESULTS SUMMARY
Model: TinyLlama-1.1B-Chat-v1.0
Hardware: Tesla T4

HUGGINGFACE RESULTS:
Latency: 4.91s
Throughput: 162.86 tokens/sec
Peak VRAM: 2.08 GB

VLLM RESULTS:
Latency: 1.88s
Throughput: 425.28 tokens/sec
Approx Peak VRAM: 5.92 GB (Pool)

IMPROVEMENT:
Throughput Increase: 2.61x
"""

with open('experiment_results.txt', 'w') as f:
    f.write(results_text)

from google.colab import files
files.download('experiment_results.txt')
```

